Expectation of a Random Variable

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Introduction

The **expectation** $E[X]$ is the probability-weighted average of the values a random variable can take. It is the centre of gravity of the distribution, the long-run average of repeated independent draws, and the first moment in the moment hierarchy. Along with variance, it is the single most commonly reported summary of a distribution.

Prerequisites

Random variables, PMF or PDF, basic summation and integration.

Theory

For a discrete random variable with PMF p_X:

$$E[X] = \sum_x x\, p_X(x).$$

For a continuous random variable with PDF f_X:

$$E[X] = \int x\, f_X(x)\, dx.$$

Law of the Unconscious Statistician (LOTUS): for a measurable function g,

$$E[g(X)] = \sum_x g(x) p_X(x) \quad \text{or} \quad \int g(x) f_X(x)\, dx,$$

without needing to find the distribution of $g(X)$ first.

Linearity. Expectation is linear regardless of dependence:

$$E[aX + bY] = aE[X] + bE[Y], \qquad E\left[\sum_i X_i\right] = \sum_i E[X_i].$$

Product rule for independent variables: $E[XY] = E[X]E[Y]$ when $X \perp Y$.

Monotonicity: $X \leq Y$ (pointwise) $\Rightarrow E[X] \leq E[Y]$.

Expectation need not exist. The Cauchy distribution has no finite mean because the integral diverges. Existence requires $E[|X|] < \infty$.

Assumptions

- $E[|X|] < \infty$ for the expectation to be well-defined.
- Fubini/Tonelli conditions for interchanging integration and summation when needed (automatic for non-negative variables and for integrable ones).

R Implementation

```r
set.seed(2026)

# Exact expectation for discrete RV: binomial
n <- 10; p <- 0.3
k <- 0:n
E_X <- sum(k * dbinom(k, n, p))
c(formula = E_X, closed_form = n * p)

# Exact expectation for continuous RV: gamma
integrate(function(x) x * dgamma(x, shape = 3, rate = 0.5), 0, Inf)
3 / 0.5   # closed form alpha/beta

# LOTUS: E[g(X)] without the distribution of g(X)
# For X ~ Gamma(3, 0.5), compute E[log(X)]
integrate(function(x) log(x) * dgamma(x, shape = 3, rate = 0.5), 0, Inf)

# Monte Carlo estimate (LLN justifies this)
X <- rgamma(1e5, shape = 3, rate = 0.5)
mean(log(X))

# Linearity check: E[2X + 3Y] = 2E[X] + 3E[Y] regardless of dependence
X <- rnorm(1e5, 5, 2)
Y <- 0.7 * X + rnorm(1e5, 0, 1)        # dependent
c(direct = mean(2 * X + 3 * Y),
  sum_of_parts = 2 * mean(X) + 3 * mean(Y))
```

Output & Results

```
   formula  closed_form
       3.0          3.0

6 with absolute error < 8.5e-05

[1] 1.506         # E[log X] via integration
[1] 1.505         # same via Monte Carlo

     direct  sum_of_parts
     27.43         27.43
```

Closed-form, numerical integration, and Monte Carlo all agree, and linearity of expectation holds despite dependence between X and Y.

Interpretation

When a paper reports a mean, it is (implicitly) reporting an estimate of the expectation of a random variable. The sample mean converges to the expectation as n grows (law of large numbers), so inferences about population-level expectations derive their validity from this convergence.

Practical Tips

- Linearity is unconditional on dependence; this is the single most useful fact about expectation. It underlies the unbiasedness of the sample mean regardless of the correlation structure.
- The product rule requires independence: $E[XY] = E[X]E[Y]$ fails when X, Y are dependent.
- Jensen's inequality: $E[g(X)] \geq g(E[X])$ for convex g (reversed for concave). Taking a log of an expectation is not the expectation of the log.
- For heavy-tailed distributions (Cauchy, stable with index < 1), the mean may not exist; rely on medians instead.
- Monte Carlo estimation of expectations works for any integrable X; convergence rate is $O(n^{-1/2})$ (CLT).

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Probability, conditional probability, Bayes' theorem
- Discrete distributions: Bernoulli, binomial, Poisson
- Continuous distributions and Q-Q plots